

Translated

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Floyd-Warshall Algorithm

Given a directed or an undirected weighted graph G with n vertices. The task is to find the length of the shortest path d_{ij} between each pair of vertices i and j .

The graph may have negative weight edges, but no negative weight cycles.

If there is such a negative cycle, you can just traverse this cycle over and over, in each iteration making the cost of the path smaller. So you can make certain paths arbitrarily small, or in other words that shortest path is undefined. That automatically means that an undirected graph cannot have any negative weight edges, as such an edge forms already a negative cycle as you can move back and forth along that edge as long as you like.

This algorithm can also be used to detect the presence of negative cycles. The graph has a negative cycle if at the end of the algorithm, the distance from a vertex v to itself is negative.

This algorithm has been simultaneously published in articles by Robert Floyd and Stephen Warshall in 1962. However, in 1959, Bernard Roy published essentially the same algorithm, but its publication went unnoticed.

Description of the algorithm

The key idea of the algorithm is to partition the process of finding the shortest path between any two vertices to several incremental phases.

Let us number the vertices starting from 1 to n . The matrix of distances is $d[i][j]$.

Before k -th phase ($k = 1 \dots n$), $d[i][j]$ for any vertices i and j stores the length of the shortest path between the vertex i and vertex j , which contains only the vertices $\{1, 2, \dots, k-1\}$ as internal vertices in the path.

In other words, before k -th phase the value of $d[i][j]$ is equal to the length of the shortest path from vertex i to the vertex j , if this path is allowed to enter only the vertex with numbers smaller than k (the beginning and end of the path are not restricted by this property).

It is easy to make sure that this property holds for the first phase. For $k = 0$, we can fill matrix with $d[i][j] = w_{ij}$ if there exists an edge between i and j with weight w_{ij} and $d[i][i]$